Course 6024C | Semester VI | Theory | 4 Credits

Production Technology

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Mechanical Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6024C as Production Technology in Semester VI for Mechanical Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Kharagpur’s Manufacturing Processes II course and polytechnic production technology curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Tool designs, machining parameters, press operations and production plans must be based on approved engineering designs, certified equipment specifications, current safety standards and competent professional review.

Verified source note: Verified sources support broaching, grinding, superfinishing, screw thread production, gear manufacturing, jigs and fixtures, and non-traditional manufacturing (USM, WJM, ECM, EDM, EBM, LBM). The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Production Technology studies the advanced manufacturing processes, tool design and non-traditional machining methods used in modern industry. It develops the ability to plan gear and thread manufacturing, design jigs and fixtures, evaluate unconventional machining processes (EDM, LBM, WJM), and understand press tool operations while respecting the technical and safety boundaries of production engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□.

Prerequisite foundation

Manufacturing processes, engineering materials, machine drawing, workshop technology and basic structural mechanics.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain advanced machining processes, including broaching, grinding, superfinishing, gear and thread manufacturing.
2. Explain the principles and design of jigs and fixtures for precise workpiece location and clamping in machine shops.
3. Explain non-traditional manufacturing processes, including ultrasonic, water jet, electro-chemical and laser beam machining.
4. Explain press tool operations, types of dies, and the principles of shearing action in sheet metal manufacturing.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□□ □□□□□□ □□□□□ □□□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain advanced machining operations and the production of gears and screw threads.	Understand
CO2	Explain the principles of location and clamping for the design of jigs and fixtures.	Understand
CO3	Explain the working principles and applications of non-traditional manufacturing processes.	Understand
CO4	Explain press tool components, die types and the mechanics of shearing in press work.	Understand

Official CO-PO articulation matrix							
CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7-PO11**
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Advanced Machining and Gear Production	Broaching principles and applications; grinding wheels and abrasive processes; superfinishing (honing, lapping); production of screw threads (milling, rolling); gear manufacturing (hobbing, shaping).	Approx. 14 hours	CO1	Module 1
2	Jigs and Fixtures Design	Purpose of jigs and fixtures; principles of location (3-2-1 principle); locating and clamping devices; types of jigs (plate, box, channel); types of fixtures (milling, turning, welding).	Approx. 16 hours	CO2	Module 2
3	Non-Traditional Manufacturing Processes	Introduction to unconventional machining; Ultrasonic Machining (USM); Water Jet Machining (WJM); Electro-Chemical Machining (ECM); Electro-Discharge Machining (EDM); Laser Beam Machining (LBM).	Approx. 16 hours	CO3	Module 3
4	Press Tools and Die Operations	Press operations (blanking, piercing, bending, drawing); components of a press tool (punch, die, stripper); types of dies (progressive, compound, combination); shearing action and clearance.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Advanced Machining and Gear Production

Broaching principles and applications; grinding wheels and abrasive processes; superfinishing (honing, lapping); production of screw threads (milling, rolling); gear manufacturing (hobbing, shaping).

CO1 · Approx. 14 hours

Jigs and Fixtures Design

Purpose of jigs and fixtures; principles of location (3-2-1 principle); locating and clamping devices; types of jigs (plate, box, channel); types of fixtures (milling, turning, welding).

CO2 · Approx. 16 hours

Non-Traditional Manufacturing Processes

Introduction to unconventional machining; Ultrasonic Machining (USM); Water Jet Machining (WJM); Electro-Chemical Machining (ECM); Electro-Discharge Machining (EDM); Laser Beam Machining (LBM).

CO3 · Approx. 16 hours

Press Tools and Die Operations

Press operations (blanking, piercing, bending, drawing); components of a press tool (punch, die, stripper); types of dies (progressive, compound, combination); shearing action and clearance.

CO4 · Approx. 14 hours

MODULE 1

Advanced Machining and Gear Production

Broaching principles and applications; grinding wheels and abrasive processes; superfinishing (honing, lapping); production of screw threads (milling, rolling); gear manufacturing (hobbing, shaping).

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

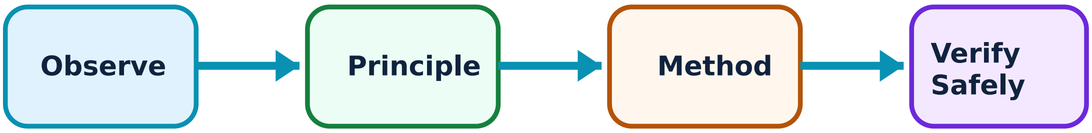
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Advanced Machining and Gear Production: identify the system, state its principle, follow the correct method and verify the result safely.

1. Grinding and superfinishing−

Grinding uses abrasive wheels to remove material for high precision and surface finish. Superfinishing processes like honing (for internal cylinders) and lapping (for flat surfaces) achieve sub-micron finishes, essential for high-performance mechanical components.

Study and application method

List the main components of a grinding wheel and compare the applications of honing and lapping in a conceptual table.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the main components of a grinding wheel and compare the applications of honing and lapping in a conceptual table.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Grinding involves high-speed rotations and dust. Do not operate grinding machines without authorised eye protection and dust extraction systems.

2. Gear and thread manufacturing−

Gears are produced by shaping (reciprocating tool) or hobbing (continuous rotating tool). Threads are produced by milling or rolling (deformation). Selection depends on the required accuracy, material and production volume.

Study and application method

Sketch the process of gear hobbing and explain the advantages of thread rolling over thread cutting.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

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Common mistake / precaution: Gear and thread geometry must strictly follow authorised standards (e.g., AGMA/ISO) for proper mating and load distribution.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Jigs and Fixtures Design

Purpose of jigs and fixtures; principles of location (3-2-1 principle); locating and clamping devices; types of jigs (plate, box, channel); types of fixtures (milling, turning, welding).

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

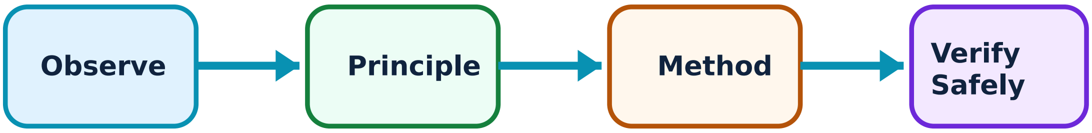
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Jigs and Fixtures Design: identify the system, state its principle, follow the correct method and verify the result safely.

1. Principles of location and clamping−

Jigs guide the tool and hold the work, while fixtures only hold the work. The 3-2-1 principle uses six points of contact to constrain all twelve degrees of freedom of a workpiece, ensuring repeatable precision in machining.

Study and application method

Explain the '3-2-1 principle' of location using a conceptual diagram; list three types of clamping devices used in fixtures.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the '3-2-1 principle' of location using a conceptual diagram; list three types of clamping devices used in fixtures.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Inadequate clamping can lead to workpiece displacement or tool breakage. Do not use fixtures that have not been authorised for the specific machining forces involved.

2. Types of jigs and fixtures−

Plate jigs are used for simple hole patterns. Box jigs enclose the workpiece for drilling on multiple faces. Fixtures are specialized for operations like milling (resisting horizontal forces) or turning (balancing rotating masses).

Study and application method

Compare a 'drill jig' and a 'milling fixture' based on their functions; describe the features of a box-type jig.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare a 'drill jig' and a 'milling fixture' based on their functions; describe the features of a box-type jig.

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Common mistake / precaution: Jigs and fixtures must be regularly inspected for wear and alignment. Worn locating pins can lead to out-of-tolerance production.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Non-Traditional Manufacturing Processes

Introduction to unconventional machining; Ultrasonic Machining (USM); Water Jet Machining (WJM); Electro-Chemical Machining (ECM); Electro-Discharge Machining (EDM); Laser Beam Machining (LBM).

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

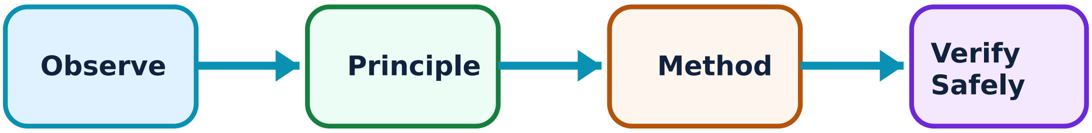
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Non-Traditional Manufacturing Processes: identify the system, state its principle, follow the correct method and verify the result safely.

1. Mechanical and chemical processes−

USM uses ultrasonic vibrations and abrasive slurry for hard/brittle materials. WJM uses high-pressure water for soft materials. ECM removes material through anodic dissolution, providing high finish without tool wear or thermal stress.

Study and application method

Describe the working principle of Ultrasonic Machining (USM); explain why ECM does not cause tool wear.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the working principle of Ultrasonic Machining (USM); explain why ECM does not cause tool wear.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Non-traditional processes involve high pressures, voltages or chemicals. Do not operate these systems without authorised training and safety equipment.

2. Thermal and electrical processes−

EDM removes material through spark erosion in a dielectric fluid, suitable for complex shapes in hard metals. LBM uses a high-intensity laser for precise cutting and drilling, involving high thermal energy in a localized zone.

Study and application method

Compare EDM and LBM based on their material removal mechanisms; list three industrial applications of Laser Beam Machining.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare EDM and LBM based on their material removal mechanisms; list three industrial applications of Laser Beam Machining.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Laser and electrical discharge processes involve radiation and electrical hazards. All operations must be conducted in authorised, shielded environments.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Press Tools and Die Operations

Press operations (blanking, piercing, bending, drawing); components of a press tool (punch, die, stripper); types of dies (progressive, compound, combination); shearing action and clearance.

Approx. 14 hours | **CO4** | **Understand · Apply**

Learning goals

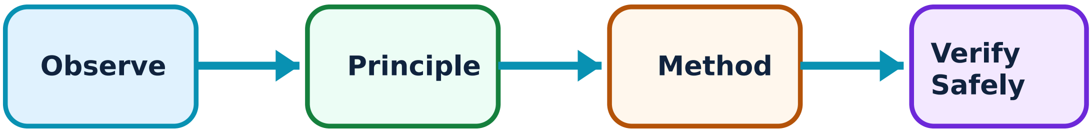
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Press Tools and Die Operations: identify the system, state its principle, follow the correct method and verify the result safely.

1. Press operations and shearing−

Press work involves deforming or cutting sheet metal using dies. Shearing action occurs when the punch penetrates the metal, involving plastic deformation and fracture. Proper clearance between punch and die is critical for clean cuts and tool life.

Study and application method

Sketch the stages of the shearing process in sheet metal; explain the difference between 'blanking' and 'piercing'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the stages of the shearing process in sheet metal; explain the difference between 'blanking' and 'piercing'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Presses exert enormous forces. Do not operate presses without authorised safety guards and 'two-hand' control systems.

2. Types of dies and components−

Progressive dies perform multiple operations at different stations as the strip moves. Compound dies perform two or more operations at a single station. Combination dies combine cutting and non-cutting (e.g., bending) operations.

Study and application method

Compare a 'progressive die' and a 'compound die' in terms of complexity and production rate; list the functions of a 'stripper plate'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare a 'progressive die' and a 'compound die' in terms of complexity and production rate; list the functions of a 'stripper plate'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Die alignment and lubrication are critical for tool life and product quality. Worn or misaligned dies can lead to catastrophic tool failure.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Advanced Machining and Gear Production

Use the concept flow to revise observation, principle, method and verification.

Module 2: Jigs and Fixtures Design

Use the concept flow to revise observation, principle, method and verification.

Module 3: Non-Traditional Manufacturing Processes

Use the concept flow to revise observation, principle, method and verification.

Module 4: Press Tools and Die Operations

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare three non-traditional machining processes in a conceptual table showing their energy source and application.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch a 3-2-1 location scheme for a rectangular block.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify the components of a simple blanking die and label them.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw a flow chart for the manufacturing of a spur gear using the hobbing process.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

List the safety equipment and protocols required for a CNC EDM machine shop.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Advanced Machining and Gear Production

Explain Grinding and superfinishing and state one correct method, application or precaution.−

Grinding uses abrasive wheels to remove material for high precision and surface finish. Superfinishing processes like honing (for internal cylinders) and lapping (for flat surfaces) achieve sub-micron finishes, essential for high-performance mechanical components.

Answer framework: List the main components of a grinding wheel and compare the applications of honing and lapping in a conceptual table.

Important point: Grinding involves high-speed rotations and dust. Do not operate grinding machines without authorised eye protection and dust extraction systems.

Explain Gear and thread manufacturing and state one correct method, application or precaution.−

Gears are produced by shaping (reciprocating tool) or hobbing (continuous rotating tool). Threads are produced by milling or rolling (deformation). Selection depends on the required accuracy, material and production volume.

Answer framework: Sketch the process of gear hobbing and explain the advantages of thread rolling over thread cutting.

Important point: Gear and thread geometry must strictly follow authorised standards (e.g., AGMA/ISO) for proper mating and load distribution.

Module 2 — Jigs and Fixtures Design

Explain Principles of location and clamping and state one correct method, application or precaution.−

Jigs guide the tool and hold the work, while fixtures only hold the work. The 3-2-1 principle uses six points of contact to constrain all twelve degrees of freedom of a workpiece, ensuring repeatable precision in machining.

Answer framework: Explain the '3-2-1 principle' of location using a conceptual diagram; list three types of clamping devices used in fixtures.

Important point: Inadequate clamping can lead to workpiece displacement or tool breakage. Do not use fixtures that have not been authorised for the specific machining forces involved.

Explain Types of jigs and fixtures and state one correct method, application or

precaution.−

Plate jigs are used for simple hole patterns. Box jigs enclose the workpiece for drilling on multiple faces. Fixtures are specialized for operations like milling (resisting horizontal forces) or turning (balancing rotating masses).

Answer framework: Compare a 'drill jig' and a 'milling fixture' based on their functions; describe the features of a box-type jig.

Important point: Jigs and fixtures must be regularly inspected for wear and alignment. Worn locating pins can lead to out-of-tolerance production.

Module 3 — Non-Traditional Manufacturing Processes

Explain Mechanical and chemical processes and state one correct method, application or precaution.−

USM uses ultrasonic vibrations and abrasive slurry for hard/brittle materials. WJM uses high-pressure water for soft materials. ECM removes material through anodic dissolution, providing high finish without tool wear or thermal stress.

Answer framework: Describe the working principle of Ultrasonic Machining (USM); explain why ECM does not cause tool wear.

Important point: Non-traditional processes involve high pressures, voltages or chemicals. Do not operate these systems without authorised training and safety equipment.

Explain Thermal and electrical processes and state one correct method, application or precaution.−

EDM removes material through spark erosion in a dielectric fluid, suitable for complex shapes in hard metals. LBM uses a high-intensity laser for precise cutting and drilling, involving high thermal energy in a localized zone.

Answer framework: Compare EDM and LBM based on their material removal mechanisms; list three industrial applications of Laser Beam Machining.

Important point: Laser and electrical discharge processes involve radiation and electrical hazards. All operations must be conducted in authorised, shielded environments.

Module 4 — Press Tools and Die Operations

Explain Press operations and shearing and state one correct method, application or precaution.−

Press work involves deforming or cutting sheet metal using dies. Shearing action occurs when the punch penetrates the metal, involving plastic deformation and fracture. Proper clearance between punch and die is critical for clean cuts and tool life.

Answer framework: Sketch the stages of the shearing process in sheet metal; explain the difference between 'blanking' and 'piercing'.

Important point: Presses exert enormous forces. Do not operate presses without authorised safety guards and 'two-hand' control systems.

Explain Types of dies and components and state one correct method, application or precaution.−

Progressive dies perform multiple operations at different stations as the strip moves. Compound dies perform two or more operations at a single station. Combination dies combine cutting and non-cutting (e.g., bending) operations.

Answer framework: Compare a 'progressive die' and a 'compound die' in terms of complexity and production rate; list the functions of a 'stripper plate'.

Important point: Die alignment and lubrication are critical for tool life and product quality. Worn or misaligned dies can lead to catastrophic tool failure.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Advanced Machining and Gear Production.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Jigs and Fixtures Design.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Non-Traditional Manufacturing Processes.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Press Tools and Die Operations.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.

2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Broaching principles and applications; grinding wheels and abrasive processes; superfinishing (honing, lapping); production of screw threads (milling, rolling); gear manufacturing (hobbing, shaping)..
2. Develop a complete answer or practical plan for: Purpose of jigs and fixtures; principles of location (3-2-1 principle); locating and clamping devices; types of jigs (plate, box, channel); types of fixtures (milling, turning, welding)..
3. Develop a complete answer or practical plan for: Introduction to unconventional machining; Ultrasonic Machining (USM); Water Jet Machining (WJM); Electro-Chemical Machining (ECM); Electro-Discharge Machining (EDM); Laser Beam Machining (LBM)..
4. Develop a complete answer or practical plan for: Press operations (blanking, piercing, bending, drawing); components of a press tool (punch, die, stripper); types of dies (progressive, compound, combination); shearing action and clearance..

LAST-MILE REVISION

Quick Revision

Module 1: Advanced Machining and Gear Production

Broaching principles and applications; grinding wheels and abrasive processes; superfinishing (honing, lapping); production of screw threads (milling, rolling); gear manufacturing (hobbing, shaping).

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Jigs and Fixtures Design

Purpose of jigs and fixtures; principles of location (3-2-1 principle); locating and clamping devices; types of jigs (plate, box, channel); types of fixtures (milling, turning, welding).

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Non-Traditional Manufacturing Processes

Introduction to unconventional machining; Ultrasonic Machining (USM); Water Jet Machining (WJM); Electro-Chemical Machining (ECM); Electro-Discharge Machining (EDM); Laser Beam Machining (LBM).

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Press Tools and Die Operations

Press operations (blanking, piercing, bending, drawing); components of a press tool (punch, die, stripper); types of dies (progressive, compound, combination); shearing action and clearance.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Grinding and superfinishing	Grinding uses abrasive wheels to remove material for high precision and surface finish.
Gear and thread manufacturing	Gears are produced by shaping (reciprocating tool) or hobbing (continuous rotating tool).
Principles of location and clamping	Jigs guide the tool and hold the work, while fixtures only hold the work.
Types of jigs and fixtures	Plate jigs are used for simple hole patterns.
Mechanical and chemical processes	USM uses ultrasonic vibrations and abrasive slurry for hard/brittle materials.
Thermal and electrical processes	EDM removes material through spark erosion in a dielectric fluid, suitable for complex shapes in hard metals.
Press operations and shearing	Press work involves deforming or cutting sheet metal using dies.
Types of dies and components	Progressive dies perform multiple operations at different stations as the strip moves.

LEARNING RESOURCES

References

Recommended production-technology references

1. A. B. Chattopadhyay, Machining and Machine Tools.
2. P. C. Sharma, A Textbook of Production Technology.
3. M. H. A. Kempster, Introduction to Jigs and Fixtures Design.
4. Donaldson, Lecain and Goold, Tool Design.

Approved public production-technology sources

- https://nptel.ac.in/courses/112105126
- https://nptel.ac.in/courses/112104301

These sources provide the verified study basis while official Course 6024C detail is unavailable; they do not replace official manufacturing specifications, certified tool designs, official safety codes or authorised production plans.